

	p.d. are in phase.
5(a)	$\lambda = \frac{hc}{E}$ $= \frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{4.53 \times 10^{-14}}$ $= 4.39 \times 10^{-12} \text{ m}$
5(b)	$p = \frac{h}{\lambda}$ $= \frac{6.63 \times 10^{-34}}{4.3907 \times 10^{-12}}$ $= 1.51 \times 10^{-22} \text{ Ns}$
5(c)	According to the principle of conservation of momentum, since there is no external force acting on the nucleus during its decay, the total momentum of the system before decay (zero since nucleus was originally stationary) must be equal to the total momentum of the nucleus and the decay products after decay. Since the γ – ray photon has a momentum after decay, the nucleus must be moving so that it has a momentum that can vectorially cancel out the momentum of the γ – ray photon. This is to ensure that the total momentum of the system is conserved.